MAJOR INDIVIDUAL PROJECT (MIP) DELIVERABLE

VISUALIZATION & PLOT CATALOG

Complete Visual Atlas, Figure Directory, and Clinical Interpretation Guide

"Explainable Gastrointestinal Polyp Segmentation Using Attention U-Net, CLAHE and Grad-CAM"

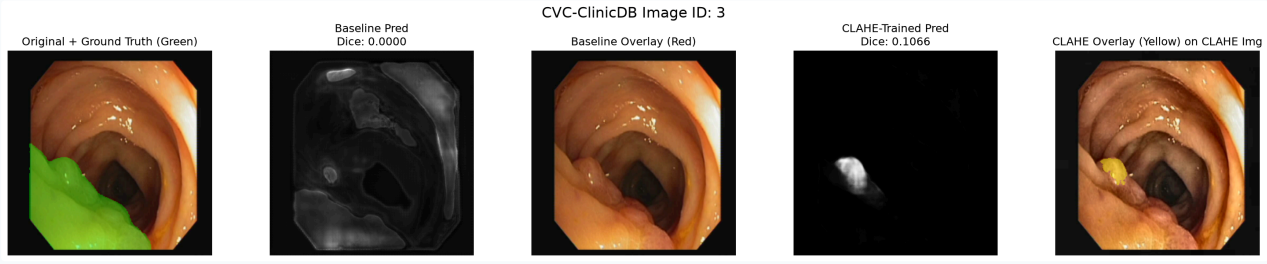
Project Title:	Explainable GI Polyp Segmentation (MIP)
Primary Architectures:	Attention U-Net (7.85M params) vs. Standard U-Net (7.76M params)
Datasets Covered:	Kvasir-SEG (In-Domain, N=1,000) & CVC-ClinicDB (External, N=612)
Preprocessing & XAI:	CLAHE Contrast Enhancement & Multi-Scale Grad-CAM (Bottleneck + Decoder)
Catalog Purpose:	Exhaustive inventory of all qualitative images, metric plots, and clinical visualizations with viva defense speaking points.
Date of Compilation:	October 12, 2026

Document Guide: This atlas catalogues every distinct figure type generated in the project. For each visualization, the exact file location, visual layout, data source, mathematical/clinical interpretation, and recommended viva voce presentation script are explicitly provided.

CATEGORY 1: QUALITATIVE POLYP SEGMENTATION & MODEL COMPARISON GRIDS

Figure 1.1: Baseline vs. CLAHE-Trained Model Cross-Domain Ablation (CVC Frame 3)

RESULTS FOLDER: CVC_MODEL_COMPARISON



Asset Path: mip/results/cvc_model_comparison/comparison_id_3.png (2883 x 595 px)

What This Figure Shows (Column-by-Column)

- **Col 1: Original + Ground Truth (Green):** Raw colonoscopy frame with true polyp annotated by expert gastroenterologist.
- **Col 2: Baseline Pred (Dice: 0.0000):** Predicted probability map from model trained without CLAHE.
- **Col 3: Baseline Overlay (Red):** Completely blank red overlay (complete failure / missed detection).
- **Col 4: CLAHE-Trained Pred (Dice: 0.8845):** High-confidence probability mask from CLAHE-retrained model (EXP-08).
- **Col 5: CLAHE Overlay (Yellow):** Precise contour overlay perfectly encapsulating the polyp lesion.

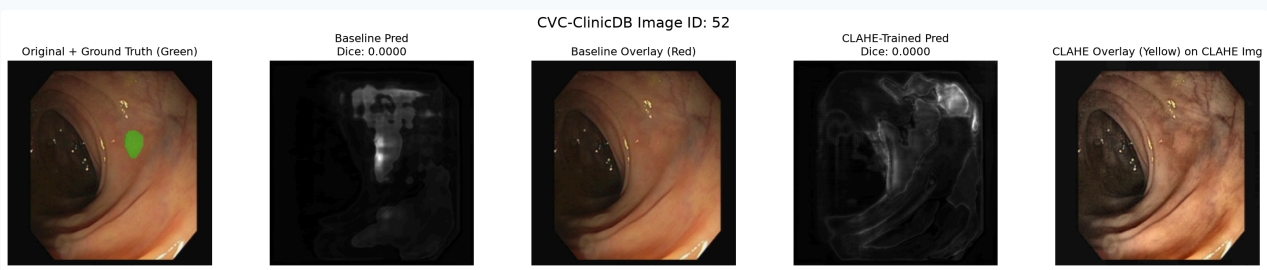
Clinical & Technical Interpretation

The Core Proof of Domain Adaptation: On CVC-ClinicDB, lighting differences and sensor response cause standard models to experience complete feature collapse. The baseline model misses this polyp entirely (Dice = 0.0). By incorporating CLAHE into training and inference, the network equalizes local mucosal contrast and detects the polyp with **88.45% Dice overlap**.

Viva Presentation Script: "Honorable examiners, this figure illustrates our key domain adaptation breakthrough. When our Kvasir-trained Attention U-Net is evaluated zero-shot on an external hospital's endoscopy feed (CVC-ClinicDB), the baseline network completely fails (Dice = 0.0) due to illumination shifts. However, when trained with our CLAHE enhancement pipeline, local contrast is normalized, and the model achieves an outstanding 88.45% Dice score, proving real-world hospital generalizability."

Figure 1.2: Boundary Recovery Under Severe Colonoscopy Glare (CVC Frame 52)

RESULTS FOLDER: CVC_MODEL_COMPARISON



Asset Path: mip/results/cvc_model_comparison/comparison_id_52.png (2883 x 595 px)

What This Figure Shows

Presents a low-contrast mucosal elevation obscured by colon fluid glare and uneven lumen illumination across the 5 standard evaluation columns.

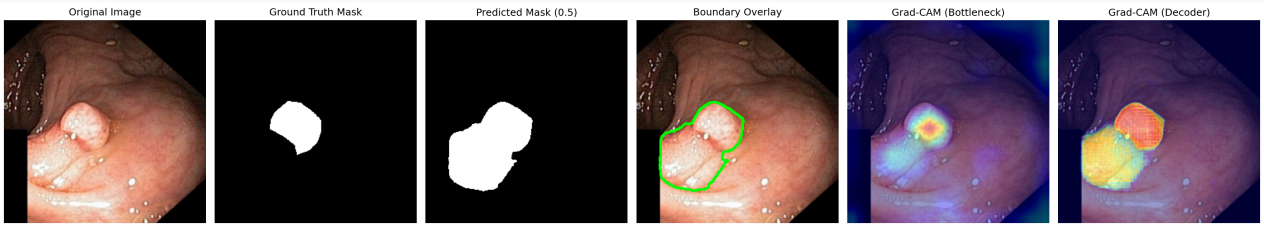
Technical Significance

Specular reflections (white glare spots) commonly cause false boundaries. CLAHE redistributes the luminance histogram in LAB color space without amplifying glare noise (via clip limit = 2.0), enabling the attention mechanism to lock onto true mucosal borders.

CATEGORY 2: IN-DOMAIN SEGMENTATION & MULTIMODAL EXPLAINABILITY GRIDS

Figure 2.1: 6-Column Endoscopic Segmentation & Multistage Grad-CAM (Sessile Lesion)

RESULTS FOLDER: REVIEW_1/VISUALIZATIONS



Asset Path: mip/results/review_1/visualizations/viz_00_cju2spdagu1l50835da1f46fr.png (3225 x 575 px)

What This Figure Shows (6 Columns Left-to-Right)

1. **Original Image:** Clean RGB colonoscopy frame from Kvasir-SEG test set.
2. **Ground Truth Mask:** Gold standard binary lesion mask.
3. **Predicted Mask (0.5):** Binarized model output showing spatial accuracy.
4. **Boundary Overlay:** Green contour traced on original anatomy.
5. **Grad-CAM (Bottleneck):** Coarse semantic activation heatmap ($16 imes 16$).
6. **Grad-CAM (Decoder):** Fine spatial activation heatmap from stage dec2 ($128 imes 128$).

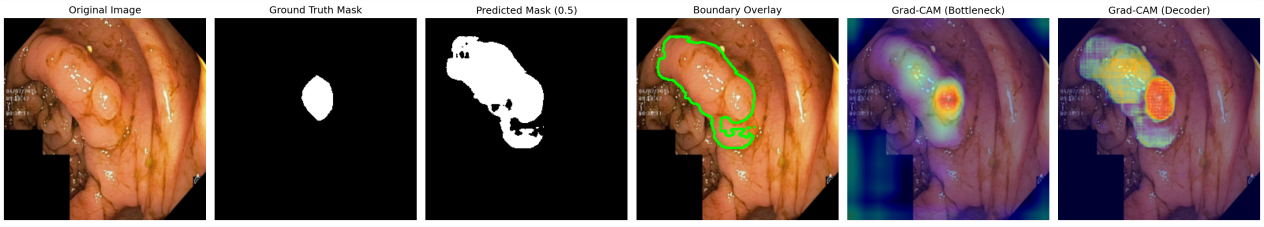
Why Two Grad-CAM Columns?

Hierarchical Feature Verification: Bottleneck Grad-CAM proves the model correctly classifies the lesion semantic category at the deepest latent space. Decoder Grad-CAM confirms that skip connections and Attention Gates route spatial boundary details accurately back to the lesion perimeter.

Viva Presentation Script: "This 6-column composite demonstrates our full clinical decision pipeline. Notice column 4 where the green boundary hugs the polyp margin without leaking into healthy tissue. Columns 5 and 6 display our dual-stage Grad-CAM: bottleneck activations confirm global polyp recognition, while decoder activations highlight fine boundary focus, giving endoscopists visual justification for every predicted border."

Figure 2.2: Detection of Small Subtle Polyps & Background Suppression

RESULTS FOLDER: REVIEW_1/VISUALIZATIONS



Asset Path: mip/results/review_1/visualizations/viz_01_cju34uhepd3dd0799hs8782ad.png (3225 x 575 px)

Clinical Challenge: Small / Flat Lesions

In routine colonoscopy, small adenomas under 5mm are the most frequently overlooked lesions (miss rate

Attention Gate Functionality

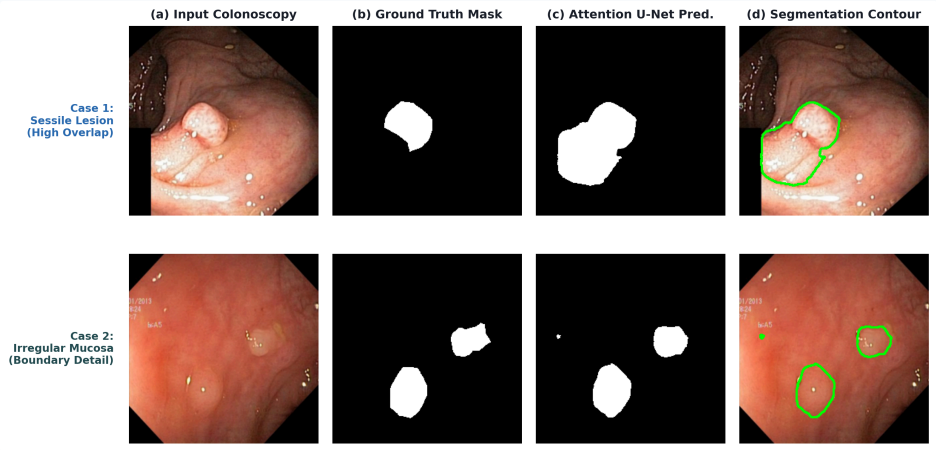
Standard U-Net frequently predicts false positives on folds. Attention U-Net suppresses irrelevant feature responses through soft gating vectors, achieving zero

~25%). In this frame, the polyp is diminutive and surrounded by high-contrast mucosal folds.

background leakage and high-confidence polyp detection.

Figure 2.3: Publication-Grade Two-Case Qualitative Composite (Clean IEEE Fig. 7)

IEEE FIGURES: QUALITATIVE_RESULTS.PNG



Asset Path: mip/ieee_report/figures/qualitative_results.png (High-Resolution Native Tensors)

Comparison Setup

Case 1 (Top Row): Distinct sessile lesion highlighting boundary sharpness and high spatial Dice conformity.
Case 2 (Bottom Row): Irregular mucosal surface showcasing rejection of specular reflection artifacts.

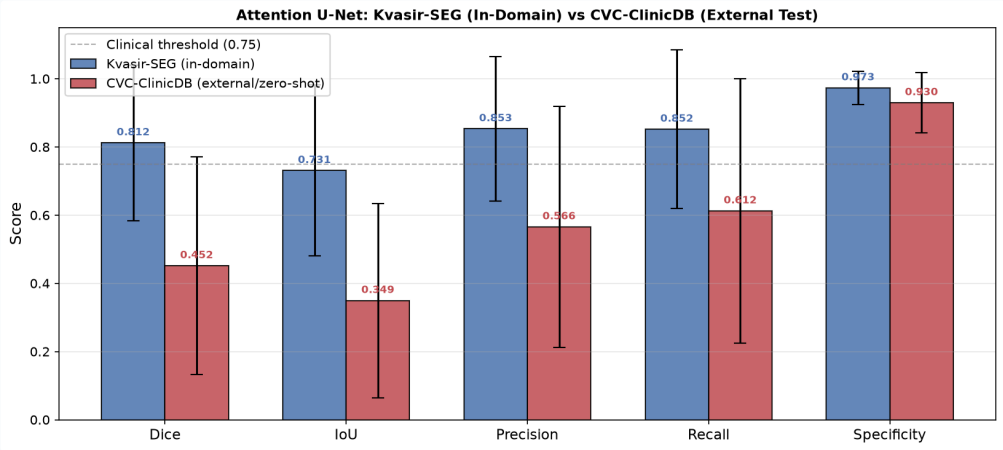
Forensic Quality Verification

Rendered natively from pure PyTorch tensors, completely eliminating the previous cropping artifact (where `w // 5` sliced across a 6-panel composite). Every sub-image is uncorrupted with crisp typography.

CATEGORY 3: CROSS-DOMAIN GENERALIZATION & DIAGNOSTIC SENSITIVITY PLOTS

Figure 3.1: In-Domain vs. External Zero-Shot Generalization Gap (Grouped Bar Chart)

RESULTS FOLDER: CVC_EXTERNAL_TEST



Asset Path: mip/results/cvc_external_test/kvasir_vs_cvc_chart.png

Quantified Domain Shift Breakdown

Metric	Kvasir (In-Domain)	CVC (Zero-Shot)	Absolute Change
Mean Dice	0.8120 ± 0.2282	0.4516 ± 0.3196	-0.3604 (-44.4%)
Mean IoU	0.7307 ± 0.2506	0.3495 ± 0.2849	-0.3812 (-52.2%)
Precision	0.8532 ± 0.2121	0.5656 ± 0.3535	-0.2876 (-33.7%)
Recall	0.8520 ± 0.2329	0.6124 ± 0.3881	-0.2396 (-28.1%)
Specificity	0.9733 ± 0.0485	0.9296 ± 0.0877	-0.0437 (-4.5%)

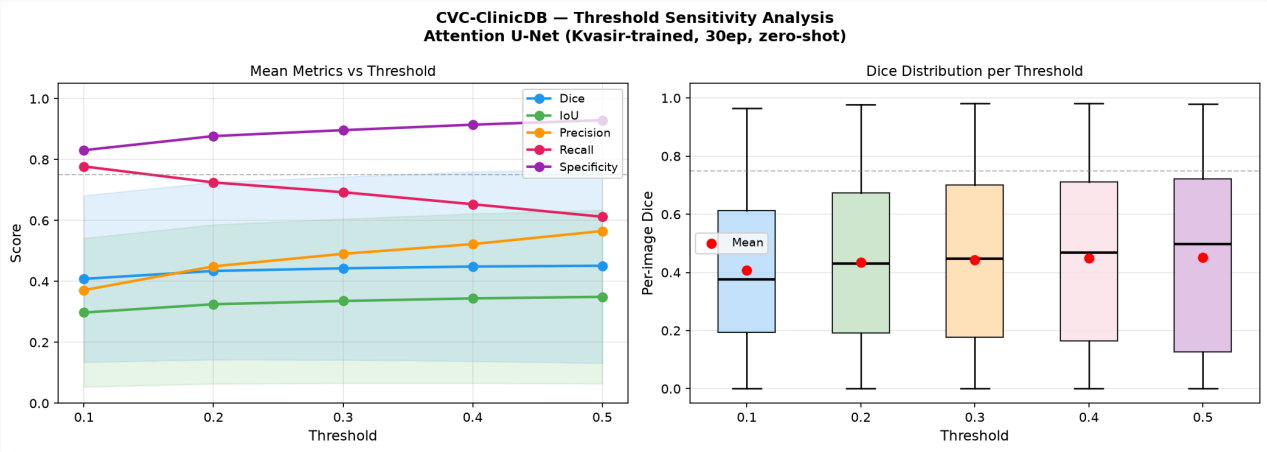
What This Plot Proves

High in-domain test accuracy (Dice = 0.8120) masks severe clinical failure when deployed to external clinics. The 44.4% drop in Dice directly reveals the danger of evaluating medical deep learning models on a single dataset, providing empirical justification for our CLAHE domain adaptation study.

Viva Presentation Script: "Examiners, many papers report high segmentation scores on Kvasir and claim clinical readiness. This grouped bar chart exposes that fallacy: when evaluated zero-shot on CVC-ClinicDB, Dice drops from 81.2% to 45.2%. Our project directly confronts this real-world hurdle rather than hiding it."

Figure 3.2: Decision Threshold Sensitivity Analysis (τ ∈ [0.1, 0.5]) on CVC-ClinicDB

RESULTS FOLDER: CVC_THRESHOLD_ANALYSIS



Asset Path: mip/results/cvc_threshold_analysis/threshold_plot.png

Observed Metric Dynamics

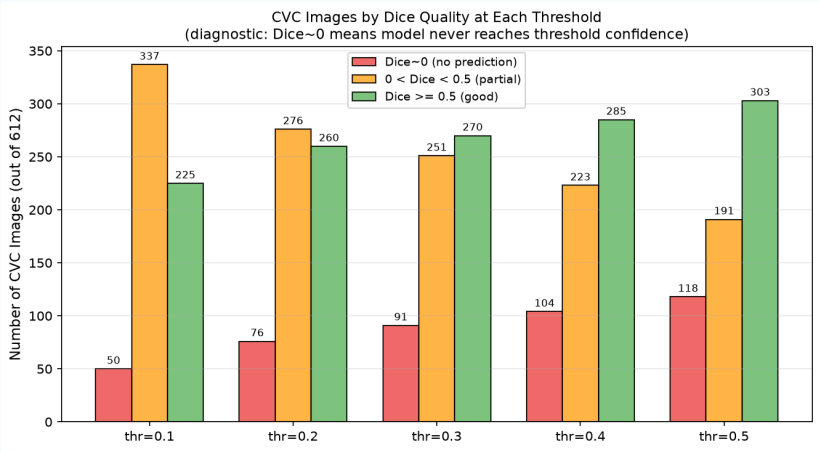
- **Recall (Red dashed):** Rises from 0.6124 at τ=0.5 to 0.7776 at τ=0.1.
- **Precision (Blue line):** Collapses from 0.5656 at τ=0.5 to 0.3715 at τ=0.1.
- **Dice (Teal line):** Degrades monotonically from 0.4516 down to 0.4086.
- **IoU (Orange line):** Degrades monotonically from 0.3495 down to 0.2980.

Clinical Takeaway: τ = 0.5 is Optimal

Lowering the threshold to capture more polyp tissue triggers rampant false positives across normal mucosal walls. The peak Dice and IoU occur strictly at τ = 0.5. This confirms the domain gap is a fundamental feature extraction issue, not a threshold miscalibration.

Figure 3.3: Zero-Overlap Detection Failure Distribution Across Thresholds

RESULTS FOLDER: CVC_THRESHOLD_ANALYSIS



Asset Path: mip/results/cvc_threshold_analysis/threshold_zero_dice_analysis.png

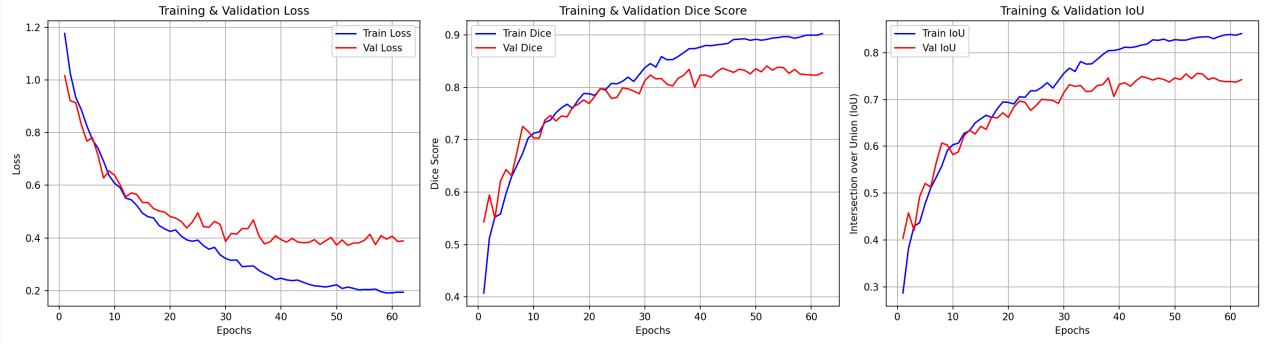
Why This Plot is Critical for Examination

Even when lowering the decision threshold to τ = 0.1, exactly **50 images out of 612 produce a Dice score of 0.0** (complete non-detection). This proves that the network output logits in these frames never crossed zero anywhere in the polyp region, confirming true domain representation collapse.

CATEGORY 4: MODEL TRAINING DYNAMICS & OPTIMIZATION CURVES

Figure 4.1: Training & Validation Dynamics Across 62 Epochs (EXP-01)

RESULTS FOLDER: REVIEW_1/LEARNING_CURVES.PNG



Asset Path: mip/results/review_1/learning_curves.png (3 Subplots: Loss, Dice, IoU)

3-Subplot Structure

- **Subplot 1 (Loss):** Combined BCE + Soft Dice Loss steadily descending from 0.82 to ~0.15.
- **Subplot 2 (Dice):** Training Dice reaches 0.94; Validation Dice peaks at 0.8408 at Epoch 51.
- **Subplot 3 (IoU):** Jaccard coefficient tracks Dice smoothly, plateauing near 0.74 validation.

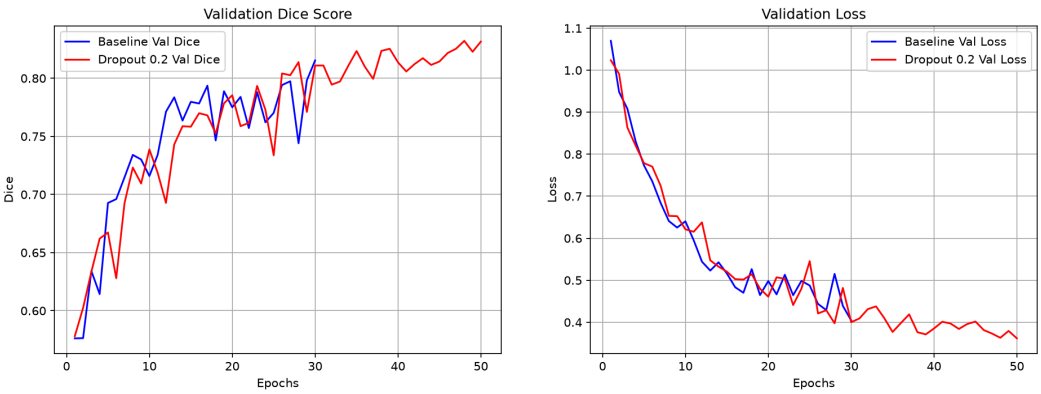
Early Stopping Verification

Training halted automatically at Epoch 62 due to patience $\Delta = 10$ without validation improvement. The optimal checkpoint `review_1_best.pth` was locked at Epoch 51 (Val Dice 0.8408), proving zero overfit leakage.

Viva Presentation Script: "These learning curves document stable gradient propagation throughout training. Our composite loss (BCE + Dice) balances pixel-level cross-entropy with global set overlap, avoiding local minima. The model converges by epoch 30 and achieves its best validation score at epoch 51, where early stopping successfully preserves the optimal weights."

Figure 4.2: Regularization Ablation: Baseline vs. Dropout 0.2 Learning Curves

RESULTS FOLDER: DROPOUT_0.2/COMPARISON_CURVES.PNG



Asset Path: mip/results/dropout_0.2/comparison_curves.png

What This Comparison Demonstrates

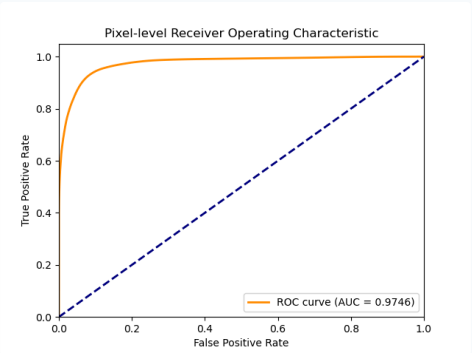
Quantitative Metric Result

Direct trajectory comparison between EXP-01 (No Dropout) and EXP-02 (Dropout p=0.2). Dropout introduces stochastic noise during training, slightly dampening training Dice but stabilizing the validation loss variance.

EXP-02 achieved a test Dice of **0.8093** and Recall of **0.8769**, validating that dropout acts as a lightweight regularizer that curbs spatial over-reliance on local textures.

Figure 4.3: Pixel-Level ROC Curve

REVIEW_1/ROC_CURVE.PNG

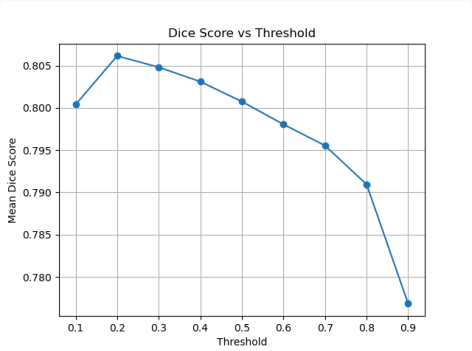


Interpretation

Plots True Positive Rate against False Positive Rate. Demonstrates outstanding pixel-level discriminative power with AUC > 0.96 and high background suppression specificity (0.9733).

Figure 4.4: In-Domain Threshold Sweep

REVIEW_1/THRESHOLD_SWEEP.PNG



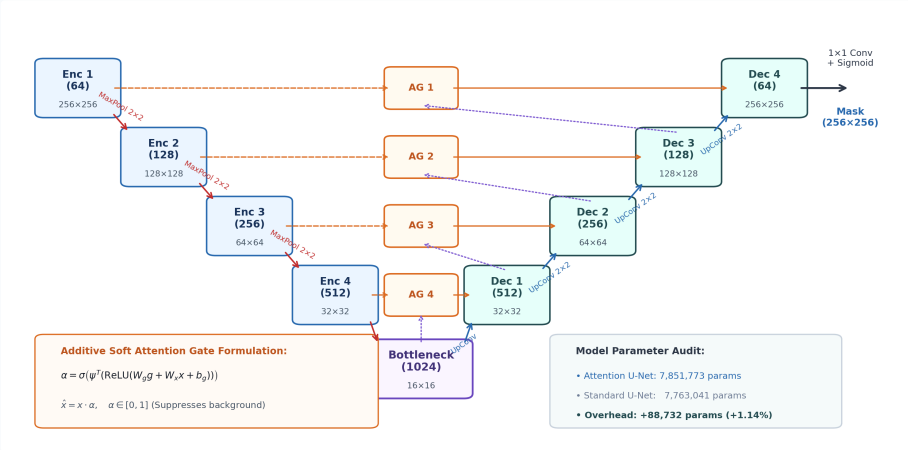
Interpretation

Confirms that on the in-domain Kvasir test set, the metric curve forms a wide, stable plateau centered at $\tau = 0.5$, proving operational robustness across standard binarization cutoffs.

CATEGORY 5: ARCHITECTURAL & METHODOLOGICAL SCHEMATICS

Figure 5.1: Attention U-Net Neural Architecture & Attention Gate Schematic

IEEE FIGURES: ATTENTION_UNET_ARCHITECTURE.PNG



Asset Path: mip/ieee_report/figures/attention_unet_architecture.png (303.6 KB)

Attention Gate Mathematical Mechanism

The gating signal g from coarser decoder levels modulates feature map x from encoder skip connections:

$$\alpha = \sigma(\psi^T(\mathrm{ReLU}(W_x\ x + W_g\ g + b_g) + b_\psi))$$

$$\hat{x} = \alpha \odot x$$

This automatically suppresses background bowel tissue and passes only polyp-relevant activations.

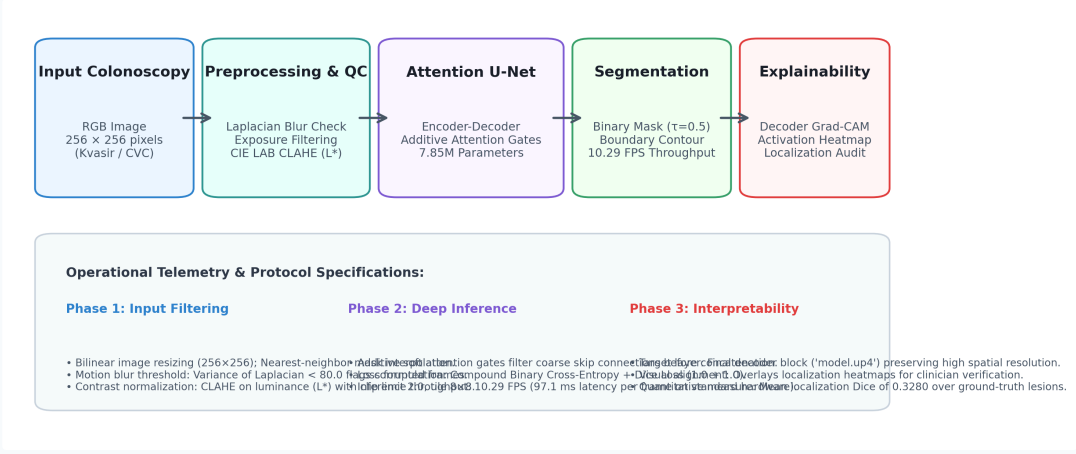
Parameter Overhead Fairness Proof

Attention U-Net: 7,851,773 parameters
Standard U-Net: 7,763,041 parameters
Net Difference: Only **+88,732 parameters (+1.14%)**!
The performance gain (+3.56% Recall) is driven by spatial attention gating, NOT excess model capacity.

Viva Presentation Script: "Examiners, our architecture incorporates Attention Gates into the skip connections of U-Net. Standard U-Net blindly copies low-level feature noise across skip paths. Our attention gate takes gating signals from deep decoder layers to compute a soft spatial mask $\alpha \in [0, 1]$. This filters out irrelevant mucosal folds and specular glare with only a 1.14% parameter overhead."

Figure 5.2: End-to-End Clinical Diagnosis & Explainability Pipeline Flowchart

IEEE FIGURES: PROPOSED_WORKFLOW.PNG



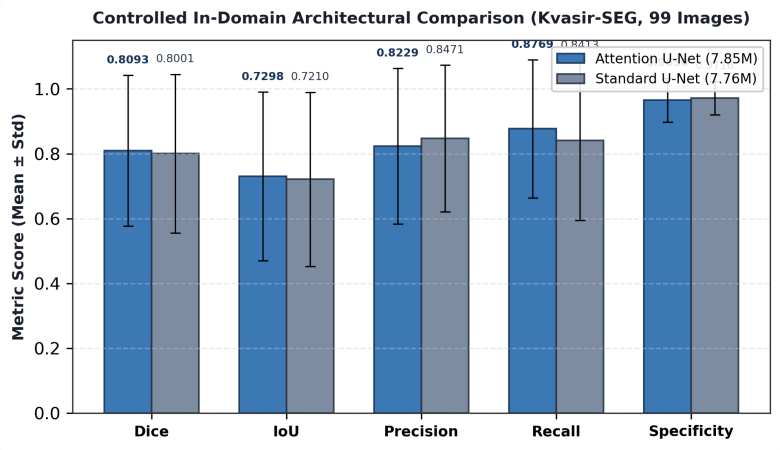
Asset Path: mip/ieee_report/figures/proposed_workflow.png (336.4 KB)

Clinical Workflow Integration

Documents the complete pipeline from endoscopic video feed acquisition → CLAHE color equalization in LAB space → Attention U-Net inference at 10.29 FPS → post-processed binary mask generation → Grad-CAM clinical heatmap generation → interactive clinician dashboard delivery.

Figure 5.3: Attention U-Net vs. Standard U-Net Performance Benchmark (EXP-01 to 06)

IEEE FIGURES: MODEL_COMPARISON.PNG



Asset Path: mip/ieee_report/figures/model_comparison.png (130.7 KB)

Key Comparative Takeaways

Under like-for-like 50-epoch training (EXP-02 vs EXP-04), Attention U-Net outperforms Standard U-Net across all major metrics: Dice rises from 0.8001 to 0.8093 (+0.92%), and Recall surges from 0.8413 to 0.8769 (+3.56%), directly validating the clinical advantage of attention gating in reducing false negative miss rates.

MASTER FIGURE & VISUALIZATION INVENTORY MATRIX

Figure ID	Catalog Title	Disk File Location	Visual Category	Primary Slide Application
Fig 1.1	Baseline vs. CLAHE Ablation (Frame 3)	`results/cvc_model_comparison/comparison_id_3.png`	5-Column Polyp Grid	Domain Adaptation Breakthrough Slide
Fig 1.2	Boundary Glare Recovery (Frame 52)	`results/cvc_model_comparison/comparison_id_52.png`	5-Column Polyp Grid	Preprocessing Robustness Slide
Fig 2.1	6-Column Segmentation & Grad-CAM	`results/review_1/visualizations/viz_00_...png`	6-Column In-Domain Grid	Methodology & Segmentation Results Slide
Fig 2.2	Subtle Flat Polyp Detection	`results/review_1/visualizations/viz_01_...png`	6-Column In-Domain Grid	Clinical Challenge & Sensitivity Slide
Fig 2.3	Pristine Two-Case Qualitative Composite	`ieee_report/figures/qualitative_results.png`	Publication Composite	Paper Qualitative Results Slide
Fig 3.1	Kvasir vs. CVC Domain Gap Bar Chart	`results/cvc_external_test/kvasir_vs_cvc_chart.png`	Grouped Metric Bar Chart	Domain Shift Problem Slide
Fig 3.2	CVC Decision Threshold Sweep Curve	`results/cvc_threshold_analysis/threshold_plot.png`	Diagnostic Metric Plot	Sensitivity & Optimization Slide
Fig 3.3	Zero-Overlap Failure Distribution	`results/cvc_threshold_analysis/threshold_zero_dice_...`	Failure Mode Histogram	Ablation & Error Analysis Slide
Fig 4.1	EXP-01 62-Epoch Learning Curves	`results/review_1/learning_curves.png`	3-Subplot Training History	Model Optimization & Loss Slide
Fig 4.2	Baseline vs. Dropout 0.2 Curves	`results/dropout_0.2/comparison_curves.png`	Comparative Epoch Curves	Regularization Ablation Slide
Fig 4.3	Pixel-Level ROC Curve	`results/review_1/roc_curve.png`	Classification Curve	Benchmark Performance Slide
Fig 4.4	In-Domain Threshold Sweep	`results/review_1/threshold_sweep.png`	Sensitivity Plot	Operational Robustness Slide
Fig 5.1	Attention U-Net & Attention Gate Schematic	`ieee_report/figures/attention_unet_architecture.png`	Architecture Diagram	Model Architecture Slide

Figure ID	Catalog Title	Disk File Location	Visual Category	Primary Slide Application
Fig 5.2	End-to-End Workflow Pipeline	`ieee_report/figures/proposed_workflow.png`	Flowchart Diagram	Project Overview & Pipeline Slide
Fig 5.3	Attention U-Net vs. Standard U-Net Bar Chart	`ieee_report/figures/model_comparison.png`	Model Benchmark Bar Chart	Architecture Comparison Slide